# Problem Set 2 - Part 1: IDEF0 Model

UAV Ground Station for the Northern Sea Route (NSR)

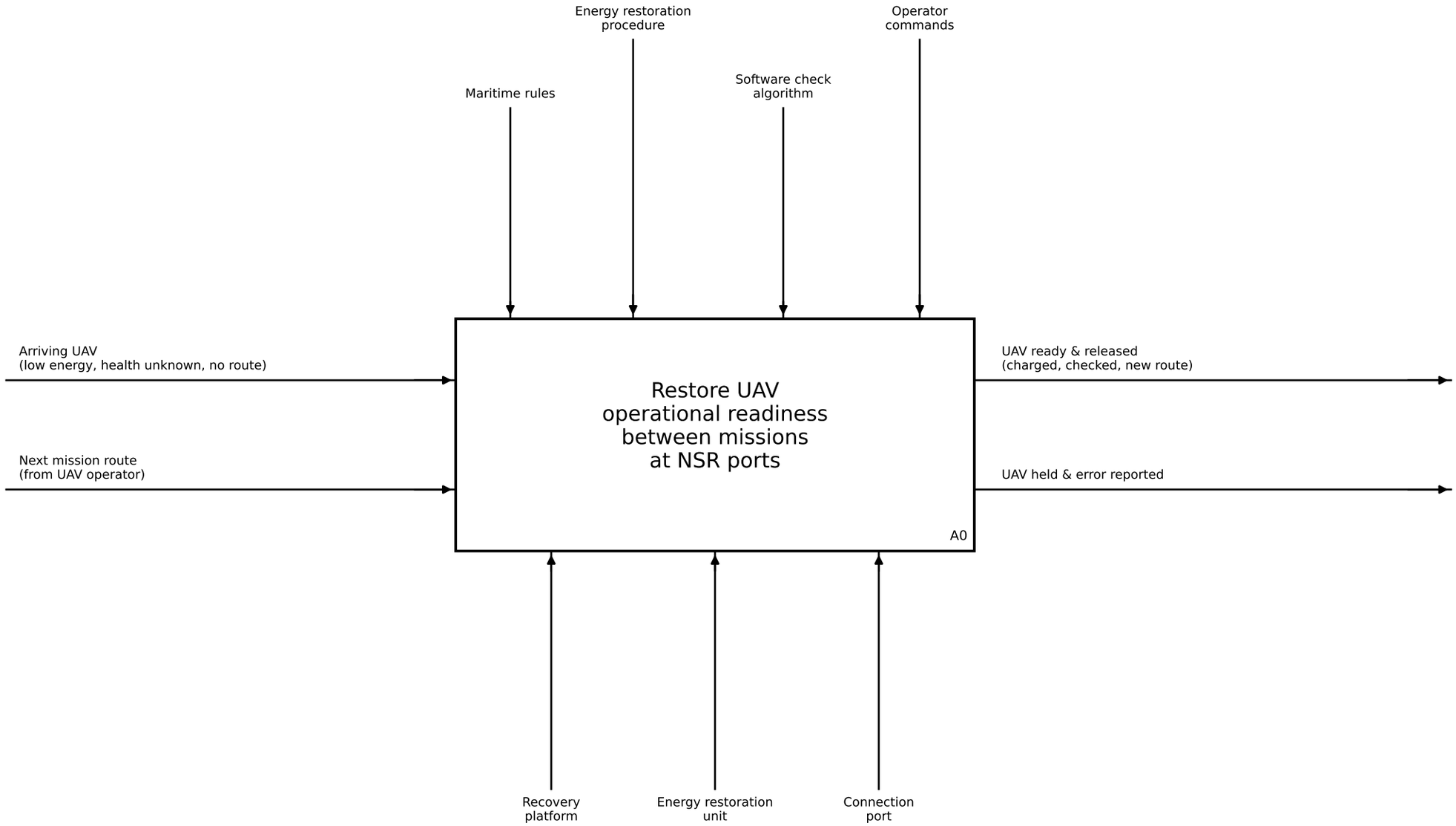
Systems Engineering 2026, Skoltech · Team 4 · 26.10.2026

| Page | Node | Diagram | Author |
|---|---|---|---|
| 1 | A-0 | Restore UAV operational readiness between missions at NSR ports (context) | Team 4 |
| 2 | A0 | Overall functions A1-A6 | Team 4 |
| 3 | A1 | Recover the UAV | Yasir Khan |
| 4 | A2 | Restore UAV energy | Varvara Vylusk |
| 5 | A3 | Check UAV health | Jad Fayad |
| 6 | A4 | Load the next route | Macmillan Gudo |

Purpose: restore UAV operational readiness between missions (restore energy, check the UAV systems and upload the next route).

Viewpoint: UGS engineering team.

| | | | | | |
|---|---|---|---|---|---|
| Used at: Skoltech | Author: Team 4<br>Project: NSR UAV Ground Station Network<br>Notes: Systems Engineering 2026 - Problem Set 2 | Date: 26.10.2026<br>Rev: 2 | ☐ WORKING<br>■ DRAFT<br>☐ RECOMMENDED<br>☐ PUBLICATION | READER DATE | CONTEXT:<br>**TOP** |

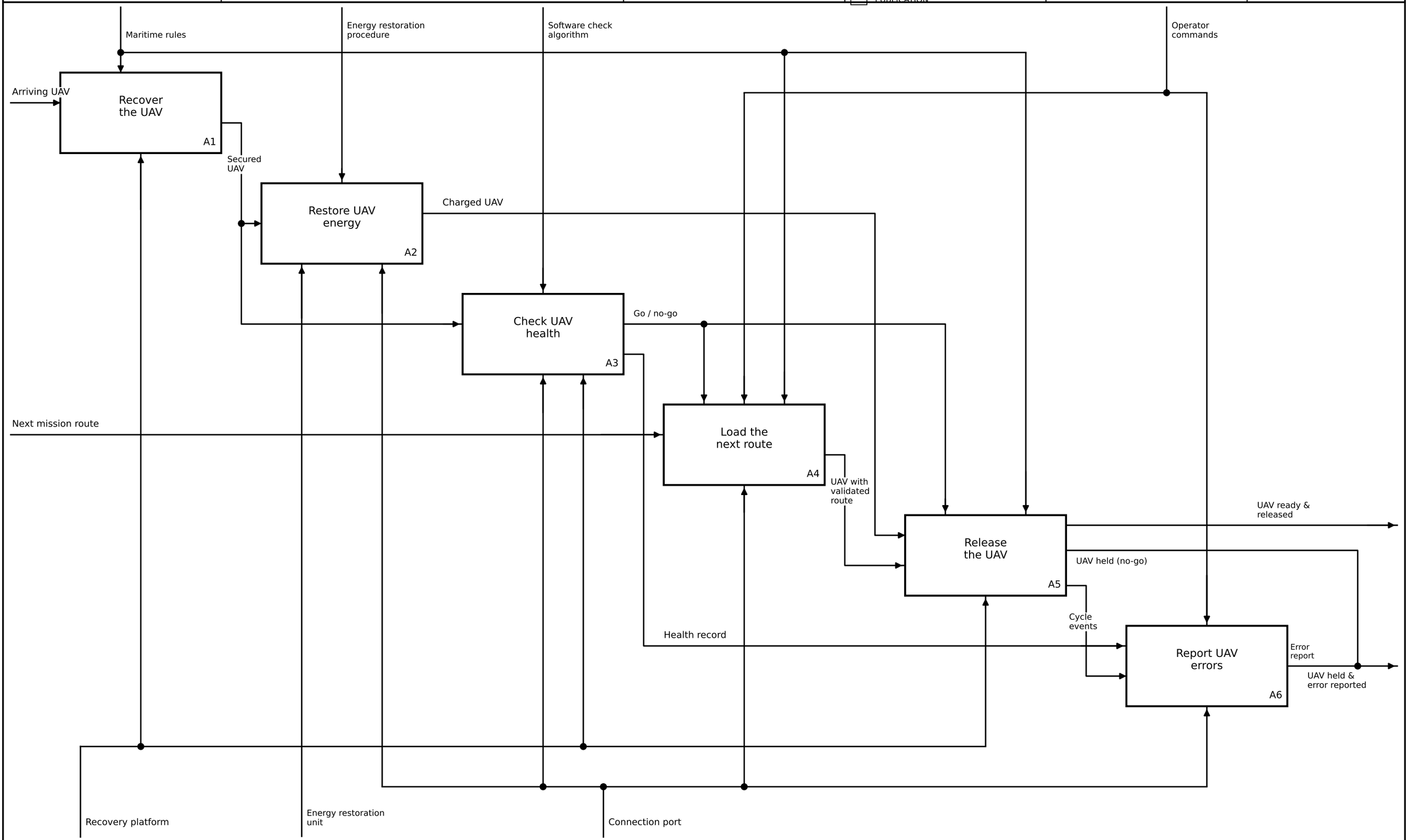



**Purpose:** Restore UAV operational readiness between missions (restore energy, check the UAV systems and upload the next route).

**Viewpoint:** UGS engineering team.

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|---|---|---|---|
| NODE: A-0 | TITLE: | Restore UAV operational readiness between missions at NSR ports | NO.: pg. 1 |

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|---|---|---|---|---|---|
| Used at: Skoltech | Author: Team 4<br>Project: NSR UAV Ground Station Network<br>Notes: Systems Engineering 2026 - Problem Set 2 | Date: 26.10.2026<br>Rev: 2 | ☐ WORKING<br>■ DRAFT<br>☐ RECOMMENDED<br>☐ PUBLICATION | READER DATE | CONTEXT: ■ |

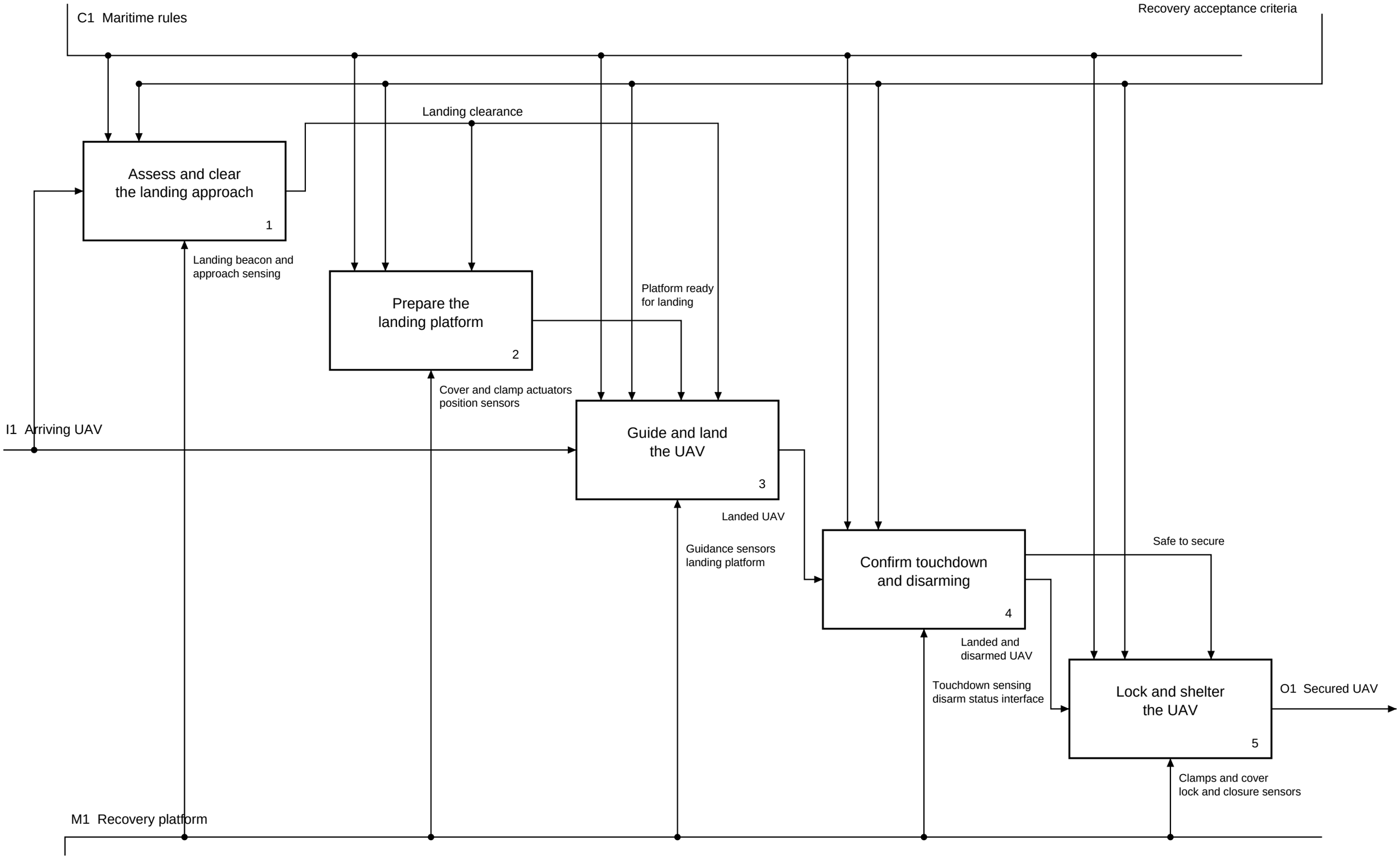



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| NODE: A0 | TITLE: | Restore UAV operational readiness between missions at NSR ports | NO.: pg. 2 |

# A1 Recover the UAV

NSR UAV Ground Station Network  Problem Set 2  Individual decomposition

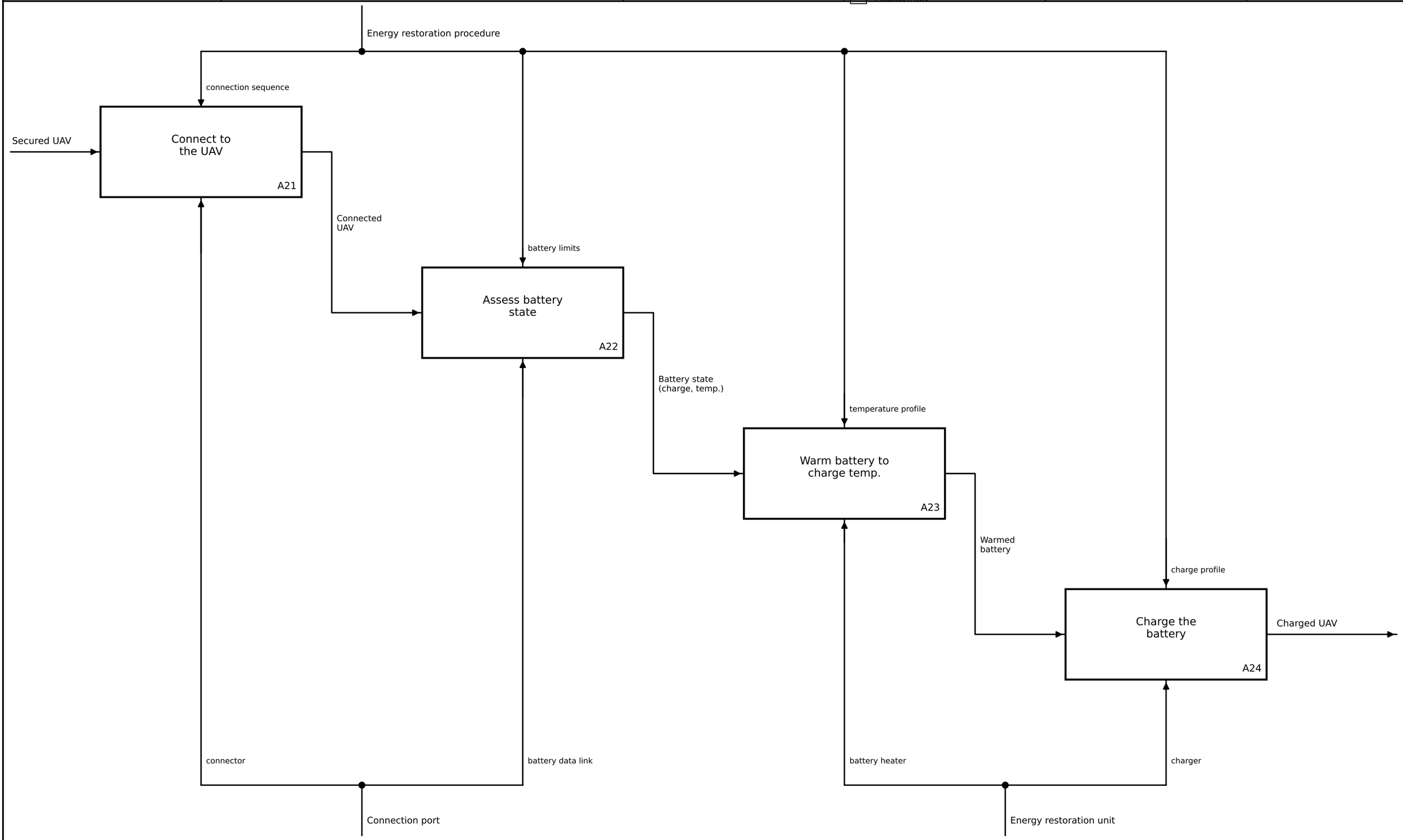
Author: Yasir Khan
Date: 26 October 2026  Revision: 3 individual proposal
Parent: A0 / function A1




NODE A1  TITLE Recover the UAV

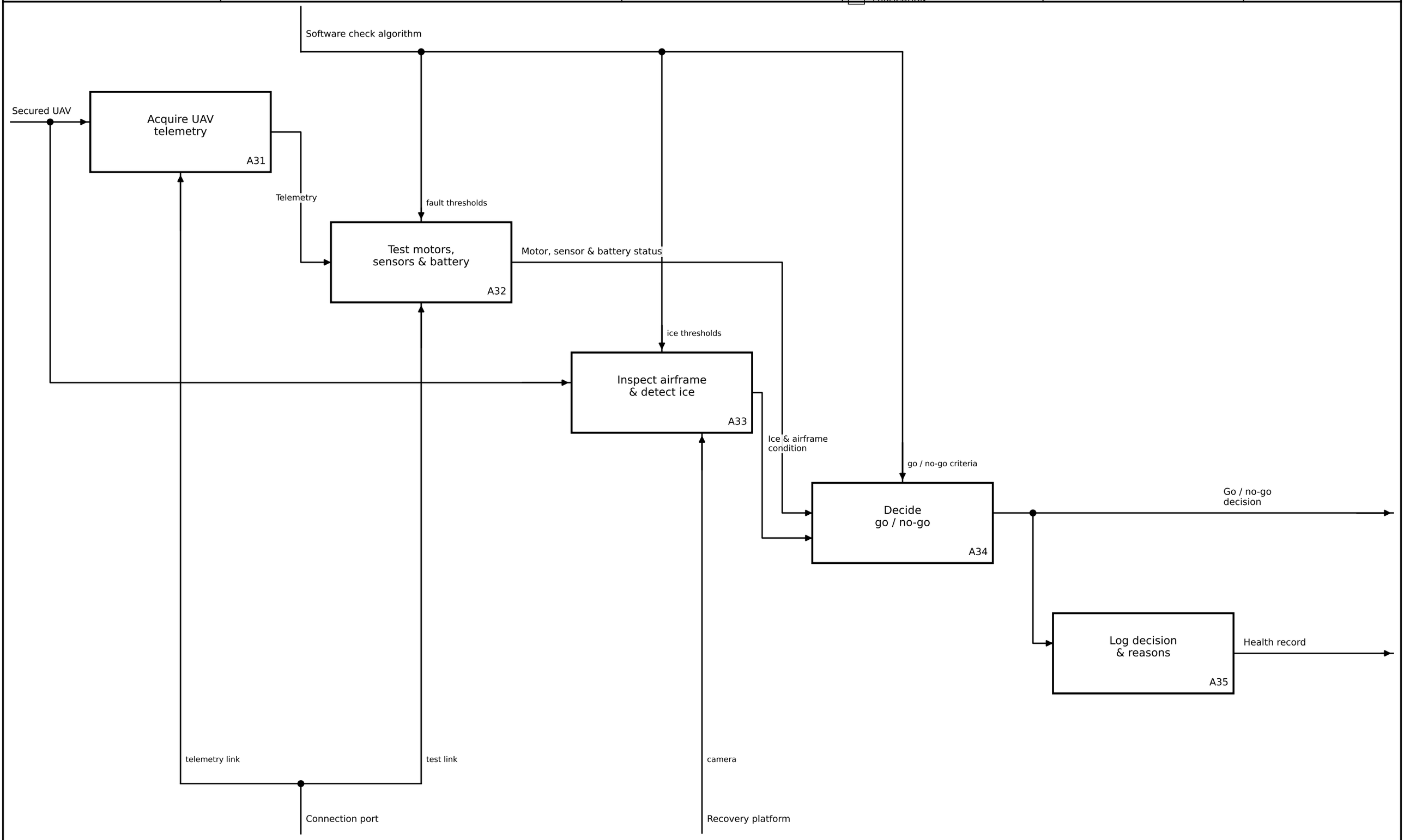
Box numbers 1-5 correspond to nodes A11-A15. Crossings without dots are not junctions.

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|---|---|---|---|---|---|
| Used at: Skoltech | Author: Varvara Vylusk<br>Project: NSR UAV Ground Station Network<br>Notes: Systems Engineering 2026 - Problem Set 2 | Date: 26.10.2026<br>Rev: 2 | ☐ WORKING<br>■ DRAFT<br>☐ RECOMMENDED<br>☐ PUBLICATION | READER DATE | CONTEXT: |



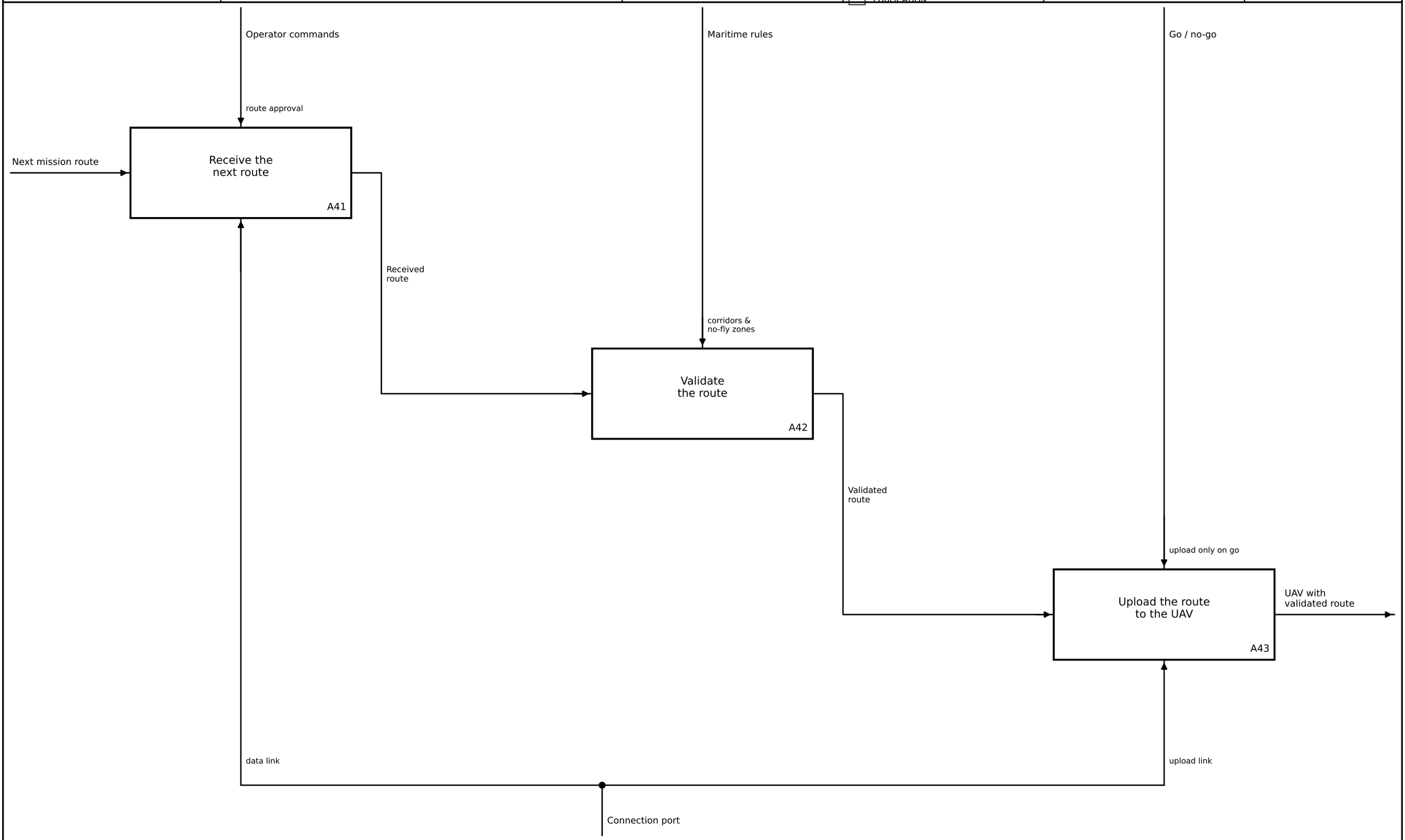

| | | | |
|---|---|---|---|
| NODE: A2 | TITLE: | Restore UAV energy | NO.: pg. 4 |

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|---|---|---|---|---|---|
| Used at: Skoltech | Author: Jad Fayad<br>Project: NSR UAV Ground Station Network<br>Notes: Systems Engineering 2026 - Problem Set 2 | Date: 26.10.2026<br>Rev: 2 | ☐ WORKING<br>■ DRAFT<br>☐ RECOMMENDED<br>☐ PUBLICATION | READER DATE | CONTEXT: |




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|---|---|---|---|
| NODE: A3 | TITLE: | Check UAV health | NO.: pg. 5 |

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|---|---|---|---|---|---|
| Used at: Skoltech | Author: Macmillan Gudo<br>Project: NSR UAV Ground Station Network<br>Notes: Systems Engineering 2026 - Problem Set 2 | Date: 26.10.2026<br>Rev: 2 | ☐ WORKING<br>■ DRAFT<br>☐ RECOMMENDED<br>☐ PUBLICATION | READER DATE | CONTEXT: |




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|---|---|---|---|
| NODE: A4 | TITLE: | Load the next route | NO.: pg. 6 |